

Intraoperative Echographic Localization of Iodine-125 Episcleral Plaque for Brachytherapy of Choroidal Melanoma

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• **PURPOSE:** To report intraoperative echographic localization of iodine-125 episcleral plaque for brachytherapy of choroidal melanoma.

• **METHODS:** In a retrospective study, 117 eyes with medium-sized choroidal melanoma in 117 patients not participating in the Collaborative Ocular Melanoma Study underwent iodine-125 episcleral plaque radiotherapy with intraoperative echographic verification of plaque placement between January 1992 and December 1998 at the Bascom Palmer Eye Institute.

• **RESULTS:** After initial plaque placement using standard localization techniques, intraoperative echography demonstrated satisfactory tumor-plaque apposition in 76% of eyes (89 of 117). In the 28 eyes (28 of 117, 24%) that required repositioning of the plaque, the extent of misplacement was less than 1 mm in 10 eyes, 1.1 to 3.0 mm in six eyes, and greater than 3 mm in eight eyes. Two eyes had tilting of the plaque, and in two additional eyes, although the plaque covered all tumor margins, the centration was considered suboptimal. Repositioning was necessary in 1 eye with an anteriorly located tumor (1 of 13, 7.7%) and in 20 eyes with peripapillary or posterior pole tumors (20 of 67, 26.3%). Anteriorly located tumors required plaque repositioning significantly less frequently than did posteriorly located tumors ($P = .041$). Misalignment involved one tumor margin in 23 eyes and two margins in five eyes. The most commonly misaligned margins were the lateral (35%) and posterior margins (26%). In no case was an anterior marginal misalignment documented. At a mean follow-up of 37 months, no tumor-related death or metastatic disease was noted. Two of the 117 patients (1.7%) had local tumor recurrence and underwent enucleation.

• **CONCLUSIONS:** Intraoperative echography is an effective adjunct for localization and confirmation of tumor-plaque relationship. This technique facilitates the identification and correction of suboptimal plaque placement at the time of surgery, potentially minimizing treatment failures. (*Am J Ophthalmol* 2000;129:199–204. © 2000 by Elsevier Science Inc. All rights reserved.)

RADIATION THERAPY IS A WELL-ESTABLISHED TREATMENT modality for uveal melanomas. Radioactive episcleral plaque therapy is the most common form of radiotherapy used^{1,2} and is associated with a 4% to 18% rate of tumor recurrence.^{3–9} Local tumor recurrence after radioactive plaque therapy is associated with an increased 5-year tumor-related mortality rate.^{4,10} A reduction in local tumor recurrence may lead to prolonged survival.

A number of factors contribute to the local tumor recurrence rate after episcleral plaque radiotherapy, including tumor resistance and subtherapeutic radiation dose delivery. Suboptimal radiation dose delivery may result from inaccurate dose calculation or plaque placement. Char and associates³ documented histologically that choroidal melanomas may extend beyond the ophthalmoscopically determined tumor basal diameter. In the prospective study by Char and associates, which used iodine-125 plaque radiotherapy, local tumor failure was decreased by extending the plaque border to at least 2 mm beyond the tumor margin.³

To ensure adequate radiation dose delivery to the entire tumor, current clinical treatment standards incorporate intraoperative plaque localization techniques coupled with a minimum 2-mm overlap margin of the plaque in relation to the underlying tumor. These techniques are particularly important when using low-energy, gamma-emitting radioisotopes, such as iodine-125, or when current plaque-shielding techniques are incorporated into the plaque design.¹¹ Low-energy isotopes are associated with the delivery of a higher radiation dose to the base of the tumor

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than is found with higher energy sources (for example, Cobalt-60), which have a faster "fall off" while maintaining a standardized dose to the tumor apex.¹²

Previously reported techniques for plaque placement include transillumination combined with indirect ophthalmoscopy to identify the tumor margins relative to the edge of the plaque,^{13,14} use of a modified fiberoptic light pipe combined with indirect ophthalmoscopy,¹⁵ postoperative magnetic resonance imaging,^{16,17} and postoperative B-scan echography.^{18,19} A recently reported technique by Finger and associates²⁰ uses a fiberoptic light affixed to the lateral plaque margin to permit direct intraoperative visualization with standard indirect ophthalmoscopy.²⁰ There are few published data concerning the use of intraoperative echography to verify plaque placement.^{8,21,22} Intraoperative imaging allows for the immediate adjustment of a suboptimally positioned plaque. The current study reports on the technique and results of intraoperative echography for placement of iodine-125 episcleral plaques.

PATIENTS AND METHODS

THE STUDY PROTOCOL WAS APPROVED BY THE INSTITUTIONAL Review Board of the University of Miami School of Medicine. The clinical records and echographic studies of consecutive patients with medium choroidal melanomas (according to Collaborative Ocular Melanoma Study criteria for medium-sized choroidal melanomas)¹¹ who underwent intraoperative echography during surgical placement of episcleral radioactive plaques at the Bascom Palmer Eye Institute were reviewed retrospectively. Patients treated within the ongoing Collaborative Ocular Melanoma Study were excluded. All procedures were performed by one surgeon (T.G.M.) between January 1992 and December 1998. Data collected included patients' demographic characteristics, tumor location and size, plaque-tumor relationship as demonstrated by intraoperative echography, need for repositioning of the plaque, and outcomes including local tumor recurrence, metastatic disease, and tumor-related death.

Episcleral plaques were applied using standard surgical techniques.^{13,14} Briefly, under sterile conditions and using retrobulbar local anesthesia, a 360-degree conjunctival peritomy was performed, and a 2-0 cotton suture was placed around the insertion of each rectus muscle. Transpupillary and transscleral (directed from the meridian opposite the tumor) transillumination were used to identify all tumor margins, which were subsequently marked on the scleral surface with a surgical pen. Tumor margins were confirmed by indirect ophthalmoscopy with scleral indentation. A 2-mm border around the tumor was then marked with calipers, and an active plaque was sewn onto the sclera using three 5-0 nylon temporary sutures.

Echographic localization of the plaque was performed either by the surgeon (T.G.M.) or a registered diagnostic

medical ultrasonographer (F.E., R.H.) using a contact B-scan instrument (Ophthascan S; Alcon Surgical, Inc, Irvine, California, or Innovative Imaging, Inc, Sacramento, California).²² To maintain sterility, the B-scan probe was placed into a sterile rubber sleeve (Roper Hall Foreign Body Locator Sterile Sheath, Model 2606-P-6078; Kieler, Broomall, Pennsylvania).²² Echographic studies were performed to view the tumor both longitudinally and transversely to verify plaque location relative to the tumor. Photographic documentation was obtained in each view, and hard copy records of plaque localization were maintained. Details of the intraoperative echography technique employed have been published previously.²²

In eyes in which the tumor-plaque apposition was satisfactory on echography, the scleral sutures were tied, maintaining the position of the plaque. In eyes that demonstrated suboptimal placement (or tilting) of the plaque on echography, sutures were adjusted and the plaque was repositioned (or the tilt was eliminated). The new plaque position was confirmed by repeating the echographic examination. At the conclusion of the surgical procedure, all plaques were found to be positioned satisfactorily, as assessed by scleral indentation combined with intraoperative echography.

A standard Collaborative Ocular Melanoma Study plaque (a silastic implant including integrated iodine-125 seed grooves placed into a gold-backed shield) with iodine-125 plaque seeds designed to deliver 85 Gy to the tumor apex was used in all cases. For tumors more than 1.5 mm from the optic nerve, the aforementioned plaque design was used and the plaque size was chosen to achieve a clear 2-mm border beyond the tumor margins. For tumors less than 1.5 mm from the optic nerve, a custom-designed notched plaque was placed adjacent to the optic nerve and centered to produce a minimum 2-mm lateral and anterior margin. In 10 consecutive patients, repeat echographic studies were performed at the time of plaque removal, and in all cases no evidence of plaque displacement was noted.

Patients were examined at 3-month to 6-month intervals after plaque placement; evaluation included a complete ophthalmologic examination, echography, and photography. Tumor recurrence was defined as an increase in tumor size (defined as greater than 0.3 mm either echographically for apical height or clinically for basal diameter) at two consecutive examinations. Patients underwent independent medical oncologic evaluations at yearly intervals.

RESULTS

ONE HUNDRED SEVENTEEN EYES WITH MEDIUM CHOROIDAL melanoma in 117 patients (66 women and 51 men) underwent iodine-125 episcleral plaque radiotherapy with intraoperative echography for plaque localization. The patients' ages ranged from 22 to 91 years (mean, 64.5; SD,

TABLE 1. Baseline Characteristics and Repositioning of Plaques in Eyes That Underwent Iodine-125 Episcleral Plaque Radiotherapy With Intraoperative Echography

	No Repositioning n = 89	Repositioned Plaques n = 28	Total n = 117	P Value
Age (years)	64.4	64.8	64.5	.59
Average tumor base diameter (mm)	9.6	9.6	9.6	.95
Tumor base–apex thickness (mm)	4.4	3.8	4.2	.12*
Tumor location				.041†
Peripapillary	9 (67%)	6 (33%)	15 (13%)	
Posterior pole	38 (73%)	14 (27%)	52 (44%)	
Equatorial	30 (81%)	7 (19%)	37 (32%)	
Anterior	12 (92%)	1 (8%)	13 (11%)	
Local recurrence	2 (2.2%)	0	2 (1.7%)	.55‡
Metastasis	1 (1.1%)	0	1 (0.9%)	1.00‡
Tumor-related death	0	0	0	N/A
Follow-up (months)	36.7	37.5	36.9	.41*

*t test.

†Exact test of trend in proportions.

‡Fisher exact test.

15.3; median, 68). The tumor location was peripapillary in 15 (13%), within the posterior pole in 52 (44%), equatorial in 37 (32%), and anterior to the equator in 13 (11%) eyes. The echographic tumor basal diameter ranged from 5.5 mm to 15.2 mm (mean, 9.6; SD, 2.5; median, 9.5), and the base-apex thickness ranged from 2.5 to 9.5 mm (mean, 4.2; SD, 1.9; median, 3.7). Baseline tumor characteristics and plaque repositioning data are summarized in Table 1. Eighty-nine eyes were found to have adequate coverage of all tumor margins with the initial plaque placement. In 28 eyes the plaque was repositioned once (23) or twice (5) to achieve satisfactory coverage of the tumor margins (Figure 1). Of the 28 eyes requiring repositioning, the extent of misplacement was less than 1 mm in 10 eyes (35.7%), 1.1 to 3.0 mm in six (21.4%), and greater than 3 mm in eight (28.6%) eyes (Table 2). Two eyes had tilting of the plaque away from the sclera, and in 2 eyes, although the plaque covered all margins, the centration was considered suboptimal. The plaque misalignment involved one tumor margin in 23 eyes and two margins in five eyes. Plaque misalignment was most commonly recognized on the lateral tumor margin (20 of 28, 71%) or the posterior tumor margin (13 of 28, 46%).

Plaque repositioning was necessary in 1 of 13 eyes (7.7%) with an anteriorly located tumor and in 20 of 76 eyes (26.3%) with a peripapillary or posteriorly located tumor. Anteriorly located tumors required plaque repositioning significantly less frequently than did posteriorly located tumors ($P = .041$). There was no significant difference in tumor size between the eyes that required plaque repositioning and those that did not require repositioning (Table 1).

The mean duration of follow-up was 37 months (range,

2 to 64; SD, 15.2; median, 24). Two patients, both of whom showed excellent plaque centration without repositioning, developed local tumor recurrence after the initial radiation therapy and subsequently underwent enucleation (2 of 117, 1.7%). No tumor-related deaths occurred.

DISCUSSION

CHOROIDAL MELANOMA IS THE MOST COMMON PRIMARY intraocular malignant neoplasm in adults. Management of patients with this neoplasm is complex and remains the subject of much discussion. The treatment modality selected is determined in part by the size and location of the tumor, and may include brachytherapy (using episcleral plaques containing iodine-125, cobalt-60, ruthenium-106, or iridium-192), charged particle therapy using protons and helium ions, photocoagulation, or laser hyperthermia therapy.^{23–28} Enucleation remains a standard in the treatment of large choroidal melanoma.

Episcleral plaque radiotherapy is frequently the treatment of choice for small- and medium-sized choroidal melanomas.^{1,2} The response to treatment is influenced by the tumor radiosensitivity, as well as the radiation dose. Effective radiation dose delivery is highly dependent on the accuracy of plaque placement relative to the tumor. Optimal plaque placement includes good centration with a 2-mm minimum border around the tumor margins, and the absence of plaque tilt.

The standard technique for plaque placement involves transillumination combined with indirect ophthalmoscopy and scleral depression to identify the tumor margins

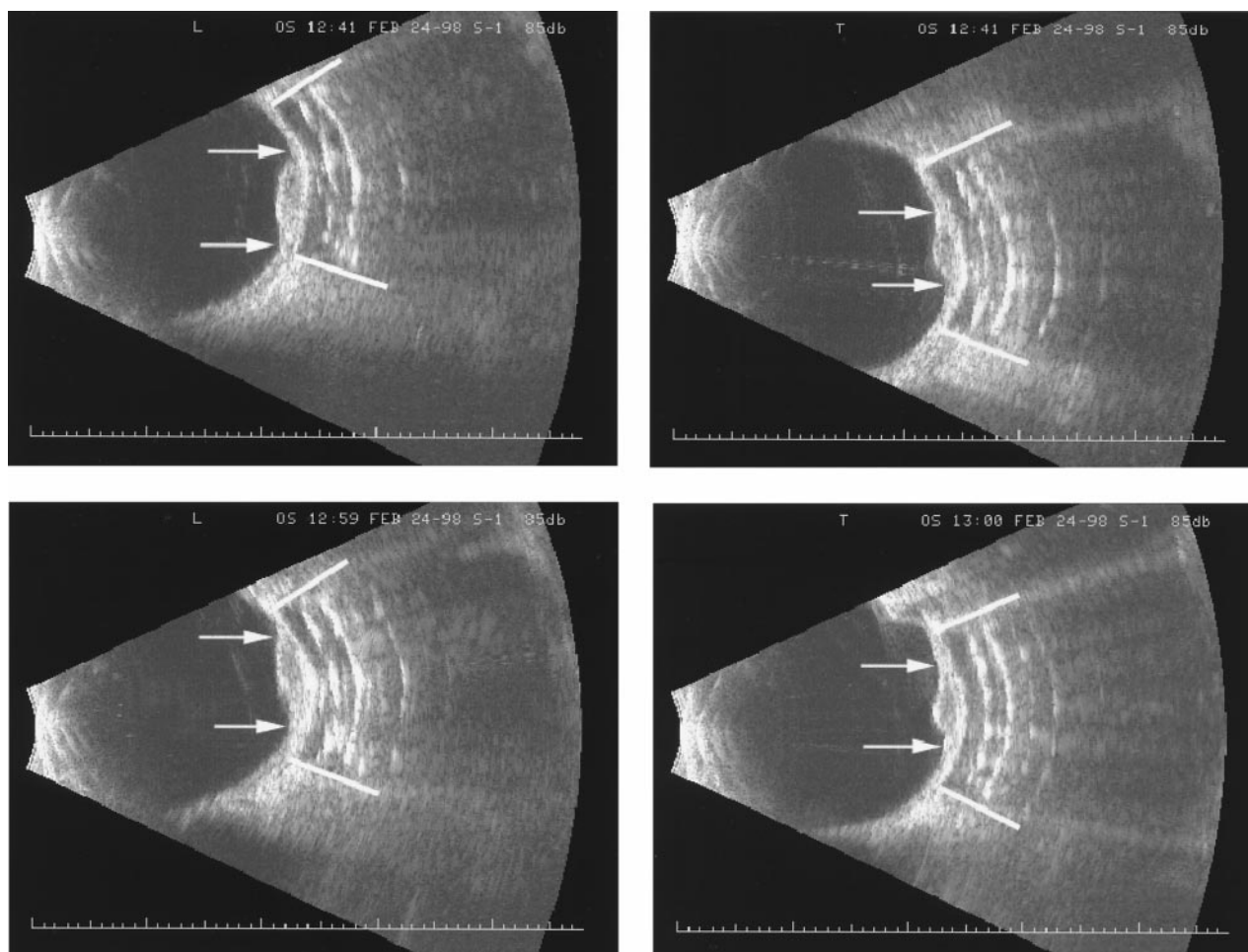


FIGURE 1. Intraoperative B-scan echograms of an episcleral plaque localized over a posterior choroidal melanoma. (Top left) Longitudinal section of an initially malpositioned plaque showing inadequate coverage of the posterior tumor margin (arrows). (Top right) Transverse section of the same patient's eye showing good coverage of the lateral tumor margins (arrows). (Bottom left) Longitudinal section of the same patient's eye after repositioning of the plaque, showing good coverage of the anterior and posterior tumor margins (arrows). (Bottom right) Transverse section of the same patient's eye after repositioning of the plaque, documenting maintenance of good coverage of the lateral tumor margins (arrows).

relative to the edge of the plaque.^{13,14} Episcleral radioactive plaques, particularly for posterior choroidal melanomas, are occasionally localized inaccurately by these standard techniques. This may be reflected in the observation that tumors located more posteriorly are associated with a higher rate of treatment failure.⁴ Other causes of treatment failure may include tumor size, location of the tumor, or histopathologic cell type.²⁹ Nonetheless, it is clear that adequate plaque positioning is important in achieving local tumor control. Various techniques have been attempted to improve the accuracy of plaque placement, including a modified fiberoptic light pipe combined with indirect ophthalmoscopy,¹⁵ postoperative magnetic resonance imaging,¹⁶ and postoperative B-scan echography.^{19,30}

Robertson and associates¹⁵ reported a series of five patients with choroidal melanoma in whom a modified

fiberoptic light pipe was used to transilluminate around the margins of the plaque, while the examiner observed the interior of the eye with indirect ophthalmoscopy, verifying the plaque position. Using this technique, two (40%) patients required readjustment of sutures after initial plaque placement. The limitation of this technique is the inability to detect any tilting of the plaque away from the sclera, particularly in posteriorly located tumors. Posterior tilting of the plaque may occur unknown to the surgeon, thus displacing the radiation source away from the tumor apex. This potential problem was addressed by Williams and associates¹⁸ in a magnetic resonance imaging study of peripapillary plaque placement in cadaver eyes. Tilting of the plaque was observed in one of six eyes. Hanna and associates¹⁶ used postoperative magnetic resonance imaging to examine tumor coverage of the episcleral plaque in 15 patients; plaques were noted to be well-centered in 12

TABLE 2. Extent of Misplacement of the Iodine-125 Episcleral Plaque Relative to the Tumor

	No. of Eyes (%)
No decentration	89 (76.1)
Suboptimal centration	2 (1.7)
<1 mm decentration	10 (8.5)
1.1–3 mm decentration	6 (5.1)
>3 mm decentration	8 (6.8)
Tilting	2 (1.7)
Total	117 (100)

eyes, eccentrically placed in two, and misplaced in one eye. B-scan echography has also been used in plaque localization. Williams and associates³⁰ confirmed accurate plaque placement in 16 patients with choroidal melanoma by performing echography on the first postoperative day. Similarly, Pavlin and associates²¹ used postoperative echography to assess the accuracy of plaque placement in nine patients with choroidal melanoma. Satisfactory echographic visualization of the tumor and plaque was obtained in all eyes. The plaque was found to be eccentrically placed in three, tilted in two, and misplaced in one eye. However, because of the postoperative nature of the examination, the misplaced plaque was not repositioned. Harbour and associates²² reported on intraoperative echographic localization of episcleral plaques in 29 patients with posterior uveal melanoma. It was noted that after initial plaque placement using standard techniques, in four eyes at least one tumor margin was not covered adequately by the plaque. Tilting of the plaque was detected in an additional two eyes. In all patients, the intraoperative nature of the echographic examination permitted immediate repositioning of the plaque.

The current study reports on the intraoperative echographic findings in 117 patients with choroidal melanoma who underwent iodine-125 episcleral plaque therapy. In 28 eyes (24%) the plaque position was adjusted in order to obtain optimal position, reconfirmed by echography. Twenty of the 28 repositioned plaques (71.4%) were in eyes with peripapillary or posterior pole tumors. Overall, plaque repositioning was necessary in 7.7% of eyes with an anteriorly located tumor and in 29.9% of eyes with peripapillary or posteriorly located tumors. Anteriorly located tumors required plaque repositioning significantly less frequently than did posteriorly located tumors ($P = .041$). There was no relationship between tumor basal diameter or tumor thickness and accurate localization of the tumor by indirect funduscopy and transillumination. In this series intraoperative echography provided rapid plaque localization and permitted immediate repositioning of all malpositioned plaques. In addition, echography provided excellent imaging of the tumor–plaque relationship in situations where ocular media were not clear, such as in

patients with vitreous hemorrhage. It also permitted detection of tilted plaques that otherwise may not have been detected. Repeat intraoperative echography performed in 10 consecutive patients did not show any plaque displacement, even in the presence of abutting extraocular muscles. In the current study using intraoperative echography, local tumor control rates were approximately 98.3%, with a mean follow-up of 37 months. In the two patients with local tumor recurrence, intraoperative repositioning was not performed, because the plaque was adequately placed. The tumors in both of these patients had a collar-button configuration, sharp lateral margins, and low reflectivity.

Intraoperative localization of choroidal melanoma may be limited in cases of small tumor size and diffuse tumor margins. Also, anterior tumors are difficult to localize using echographic techniques. In this study, we examined only choroidal melanomas that met the Collaborative Ocular Melanoma Study criteria for medium tumor size. Intraoperative echography remains an ancillary technique best used in conjunction with transillumination and indirect ophthalmoscopy to provide optimal plaque placement and to ensure adequate margins, particularly in smaller tumors or in tumors with shallow sloping margins. This complementary role is illustrated by the difficult localization of some posterior tumors with current standard clinical techniques, and yet excellent imaging of these tumors and the plaque with echographic techniques. Conversely, anterior tumors, which are difficult to localize accurately using echographic techniques, can be more easily identified with ophthalmoscopy and transillumination. The intraoperative nature of echographic examination allows for immediate repositioning of a misplaced plaque, thus eliminating the need to return to the operating room.

Relatively short follow-up after plaque treatment of these patients is an important limitation of the current study. However, intraoperative echography is a useful adjunct to the standard localization techniques in the placement of iodine-125 episcleral plaques. The procedure may be performed readily and provides rapid confirmation of the tumor–plaque relationship, thus allowing immediate repositioning of suboptimally located plaques and thereby potentially minimizing treatment failures.

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